Introduction edits — before / after

Introduction edits — before / after, for review before anything touches the Doc

Target: Google Doc "Shaked's Thesis V2" (`1YpXrDGFlzRk…`), the Introduction chapter. **Mirror:** `thesis/chapters/02_introduction.md`. Drafted 2026-08-16. **NOT YET APPLIED — nothing has touched the Doc.**

Scope = only the Introduction items Yuval explicitly raised: **email #3** (too ISFS-focused; wants sleep in general, aMCI/AD, sleep and aging), his **two margin notes with their section headings and sub-topics**, **C571** (aMCI is not only an early phase of AD, "describe also in intro"), and his **in-place tracked edits** inside the Introduction. Everything else spotted along the way is in §7 and left alone.

> **⚠ Citations carry no numbers in this document**
>
> `discussion_edits_before_after.md` is drafted and unapplied, and it renumbers the same list — six references are claimed by both passes (the C571 evidence set, which he asked for in both chapters), and it deletes two entries this pass would otherwise have numbered. Per your call on 2026-08-16, **prose and numbering are split**: citations appear here as `[@citekey]`, and a single combined numbering pass afterwards rebuilds the reference list once and writes every superscript once. That pass **supersedes `discussion_edits_before_after.md` §9**.
>
> Projected combined total: **64 entries** (30 − 2 removed + 36 unique new across both passes).

Read order: **§1 is the two decisions I need from you.** §2 is the new structure. §3–§5 are the proposed text. §6 is his tracked edits and where each one landed. §7 is what I left alone.

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§1 — DECISIONS I NEED FROM YOU

A1. What to do with the existing aging/spindles paragraph (the one he did *not* edit)

The current Introduction ¶5 begins "*Healthy aging reshapes NREM sleep and its spindles…*". Yuval left it untouched, and his heading **"Changes in sleep spindles and ISFS across aging and MCI"** sits directly above it.

The problem is created by his own request. The new §2.3 he asked for covers sleep in AD (`zhang2022alzheimerreview`, `gorgoni2016parietal`) and sleep in MCI (`taillard2019nonrem`, `liu2020spindlebiomarkers`) at length. Left as it is, ¶5 then repeats both, two paragraphs later, in compressed form.

	Option	Cost
A ✅ *recommended*	**Refocus ¶5 on what its heading says** — spindles and the ISFS. Keeps the aging-spindle sentence verbatim, rewrites the middle sentence so the parietal fast-spindle finding is used to make a *new* point (it lands on the same subtype and region as the ISFS hotspot) rather than restating §2.3, and keeps the closing biomarker sentence	Edits a paragraph he left alone

	Option	Cost
B	**Leave ¶5 exactly as it stands**	The reader meets the same two claims twice within two paragraphs; the second time adds nothing

Both versions are written out in §5 so you can compare them directly.

A2. The citation bundle [12–14] inside that paragraph

It currently bundles Champetier + Mander + Helfrich behind *"changes that themselves track cognitive decline"*. After renumbering these three are no longer contiguous, so the range has to become a list either way. While rewriting it, the accurate split is:

- Mander and Helfrich carry **the changes** (density falls, coupling loosens);
- Champetier carries **the link to cognitive decline**.

Recommended: split them. It is more accurate about which paper supports which clause, and the alternative is an unreadable three-number list. Flagged only because it touches his untouched paragraph. *(Moot if you pick B above — under B the bundle stays a bundle and just gets three numbers.)*

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§2 — the structure, and where it comes from

The §3 triage summary never listed his in-place insertions, so this was extracted directly from `Shaked's Thesis_YN.docx` (`word/document.xml`, paragraphs 62–84, walking `w:ins`). **He wrote the section headings himself.** They are reproduced verbatim below, with his sub-topic bullets:

His heading (verbatim)	His sub-topics (verbatim)
Human sleep and scalp EEG	*"Sleep what is it, how is it defined, sleep stages NREM and REM"* · *"What are the function(s) of sleep what is it good for"* · *"PSG monitoring, EEG signatures e.g. slow waves, sleep spindles etc"*
Infra-slow fluctuation of sigma power (ISFS) in NREM sleep	— (sits above the existing ISFS paragraphs)
Changes in sleep and EEG across aging and MCI	*"Sleep, Learning and Memory"* · *"Sleep and Aging"* · *"Sleep and Neurodegeneration"* · *"Alzheimer's Disease (AD)"* · *"Sleep in Alzheimer's Disease"* · *"Mild Cognitive Impairment (MCI)"* · *"Amnestic Mild Cognitive Impairment"* · *"Sleep in MCI — what is known"*
Changes in sleep spindles and ISFS across aging and MCI	— (sits above the existing aging/spindles paragraph)

His preamble to the first list — "*Put one more general paragraph to begin with – **for thesis (unnecessary for paper)***" — is the note that settled the thesis-vs-paper question.

Resulting layout, in his order, numbered to match the existing Methods `3.x` / Results `4.x`:

```
Introduction
  ¶ unheaded opener                                           kept as-is
  2.1 Human sleep and scalp EEG                               NEW,  4 ¶
  2.2 Infra-slow fluctuation of sigma power (ISFS) in NREM sleep  existing 3 ¶ + his edits
  2.3 Changes in sleep and EEG across aging and MCI           NEW,  8 ¶ (one per sub-topic)
```

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2.4 Changes in sleep spindles and ISFS across aging and MCI     existing 1 ¶ + ISFS close
2.5 The present study                                           existing 1 ¶ + his edits
```

782 words → ≈2,550. His eight sub-topics get one paragraph each; nothing is padded to hit a length.

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§3 — NEW SECTION 2.1: Human sleep and scalp EEG

BEFORE: does not exist.

AFTER:

> ### 2.1 Human sleep and scalp EEG
>
> Sleep is a reversible behavioural state, defined by reduced responsiveness to the surroundings, a species-typical posture and site, and — critically — rapid reversibility on sufficient stimulation, which is what separates it from coma and anaesthesia. It is regulated rather than merely permitted: pressure to sleep accumulates with time spent awake and dissipates during sleep, while a circadian process independently sets the times of day at which sleep is most likely to occur. The interaction of the two places a single consolidated sleep period at a particular phase of the day and shapes its internal structure [@markov2006normalsleep]. That structure has two alternating states. Non-rapid eye movement (NREM) sleep is graded into progressively deeper stages — N1, N2 and N3 — and rapid eye movement (REM) sleep is distinguished by a wake-like cortical EEG occurring alongside near-complete skeletal muscle atonia. The two alternate in cycles of roughly 90 minutes, and their balance shifts across the night: N3 dominates the early cycles, whereas REM periods lengthen towards morning.
>
> These states are distinguished by polysomnography (PSG), the simultaneous recording of the electroencephalogram (EEG) together with the electrooculogram and the submental electromyogram. The record is divided into consecutive 30 s epochs, and each epoch is assigned to a single stage according to the waveforms it contains [@silber2007visualscoring]. The criteria are explicitly electrographic rather than behavioural: N1 by the disappearance of the waking posterior alpha rhythm and the appearance of slow eye movements; N2 by sleep spindles and K-complexes; N3 by high-amplitude slow waves occupying at least a fifth of the epoch; REM by a low-voltage, mixed-frequency EEG together with rapid eye movements and the loss of muscle tone. Because the definition of N2 is the presence of spindles, any question asked about spindles is also a question about the stage in which they are scored.
>
> Why sleep is necessary is not fully settled, but several functions are well supported. The best characterised is its role in memory: material encoded during the day is reactivated during sleep and progressively integrated into long-term storage, and sleep after learning improves later retention across a wide range of tasks and species [@rasch2013memory]. Sleep also changes the physical state of the brain. The interstitial space expands during sleep and the exchange between cerebrospinal and interstitial fluid increases, so that metabolic waste — amyloid-β among it — is cleared substantially faster than during wakefulness [@xie2013clearance]. These two functions matter for what follows: the first ties sleep to cognition, the second to the proteins that accumulate in neurodegenerative disease.
>
> Two EEG signatures dominate NREM sleep. The first is the slow oscillation, a rhythm below 1 Hz in which cortical neurons alternate between a depolarised up state of sustained firing and a hyperpolarised down state of near-total silence; this alternation is what appears on the scalp as slow waves [@steriade1993slowoscillation]. The second is the sleep spindle, a brief waxing-and-waning burst of activity in the sigma frequency range, generated by the interaction between thalamic reticular and

> thalamocortical neurons [@fernandez2020spindles]. Spindles are the signature this thesis is concerned with, and the following section takes them up in detail.

Covers his sub-topics: what sleep is / how defined / NREM and REM stages (¶1) · PSG monitoring (¶2) · functions of sleep (¶3) · EEG signatures, slow waves and spindles (¶4).

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§4 — SECTION 2.2: existing ISFS paragraphs, with his edits applied

Three paragraphs, unchanged except where he edited them and where a definition now duplicates 2.1. His edits are shown **bold**; forced de-duplications are marked ⟨…⟩ and listed again in §6.2.

2.2 ¶1

BEFORE:

> Non-rapid eye movement (NREM) sleep occupies a large part of the human night, and its second stage (N2) is defined by sleep spindles: brief oscillations in the sigma band (11–16 Hz) generated by thalamocortical circuits[1–2]. Spindles are among the most reliable electrophysiological signatures of N2 and vary systematically in density and topography across individuals[3]. Two subtypes are commonly distinguished by frequency and scalp topography: slow spindles (≈11–13 Hz), maximal over frontal cortex, and fast spindles (≈13–16 Hz), maximal over centroparietal cortex[4]. Spindles are also closely tied to sleep-dependent memory consolidation[1].

AFTER:

> ⟨NREM⟩ sleep **occupies most of sleep in adults**, and **stage 2 (N2) is characterized by the intermittent occurrence of** sleep spindles: brief oscillations in the sigma band (11–16 Hz) generated by thalamocortical circuits [@fernandez2020spindles; @andrillon2011intracranial]. Spindles are among the most reliable electrophysiological signatures of N2 and vary in density and topography across individuals [@purcell2017characterizing]. Two subtypes are commonly distinguished by frequency and scalp topography: slow spindles (≈11–13 Hz), maximal over frontal cortex, and fast spindles (≈13–16 Hz), maximal over centroparietal cortex [@molle2011fastslow]. Spindles are also closely tied to **plasticity and** sleep-dependent memory consolidation [@fernandez2020spindles], **as well as with increased disconnection from the external sensory environment.**

> **Note on "occupies most of sleep in adults".** This is his wording, and it also settles Flavio #7, which had objected to the original "occupies most of the human *night*" — night includes wake, so "most" was wrong there; V2 had softened it to "a large part". His fix is better than both: NREM genuinely is most of *sleep*. Applied verbatim.

2.2 ¶2

BEFORE:

> Spindles are not distributed evenly through NREM sleep. They come in trains, and those trains recur in a slow rhythm of roughly one train every 50 seconds[5]. What fluctuates at that rate is the sigma envelope, the amplitude of sigma-band activity over time, which is modulated at a frequency of about 0.02 Hz. In rodents, this infra-slow fluctuation of sigma power (ISFS) organizes NREM sleep into alternating fragile

> and offline substates: during fragile periods sigma power is low and the animal is readily aroused, whereas during offline periods sigma power is high and the animal is relatively disconnected from sensory input and protected from awakening. This alternation is clocked by the locus coeruleus (LC) and the noradrenergic system[6–7]. The amplitude of these infra-slow noradrenergic oscillations is in turn coupled to spindle dynamics and to the restorative and mnemonic functions of sleep[8–9].

AFTER:

> Spindles are not distributed evenly through NREM sleep. They come in trains, and those trains recur in a slow rhythm of roughly one train every 50 seconds [@boutin2020spindleframework]. What fluctuates at that rate is the sigma envelope, the amplitude of sigma-band activity over time, which is modulated at a frequency of about 0.02 Hz. **In both rodents and humans**, this infra-slow fluctuation of sigma power (ISFS) organizes NREM sleep into alternating fragile and offline substates: during fragile periods sigma power is low and ⟨arousal is easily triggered⟩, whereas during offline periods sigma power is high and ⟨the sleeper is⟩ relatively disconnected from sensory input and protected from awakening. This alternation is **driven in part by** the locus coeruleus **norepinephrine (LC-NE) system** [@lecci2017infraslow; @osorioforero2021noradrenergic]. The amplitude of infra-slow noradrenergic oscillations is coupled to spindle dynamics and to the restorative and mnemonic functions of sleep [@kjaerby2022norepinephrine; @cardis2021corticoautonomic].

> **⟨...⟩ here is forced by his own edit.** He broadened "In rodents" to "In both rodents and humans", but the sentence then goes on to say "the animal is readily aroused" twice — which no longer works once humans are included. The two clauses are made species-neutral. Nothing else in the sentence changes.

2.2 ¶3

BEFORE:

> The rhythm is also present in the human EEG, where fast-sigma activity during N2 and slow-wave sleep is modulated on an infra-slow timescale, most prominently in the high sigma band and over centro-parieto-occipital regions[10]. How fast that modulation runs depends on what is measured: the same study placed the dominant rate of the sigma-power fluctuation somewhat below the 0.02 Hz described in rodents, and closer to 0.02 Hz when it tracked the occurrence of individual spindles instead. Dimitriades et al. (2024) subsequently narrowed the question, following the ISFS across development into young adulthood and quantifying it per channel as the peak frequency, spectral width, and area under the spectral peak of the sigma envelope; they linked these three measures to markers of arousal and of memory reactivation[11]. Consistent with that link, how tightly spindles cluster on this infra-slow timescale is related to how well memories are consolidated overnight[12].

AFTER: identical, except his one surviving preference —

> ...following the ISFS across development into young adulthood and quantifying the peak frequency, spectral width, and area under the spectral peak of the sigma envelope **for each EEG channel**; they linked these three measures to markers of arousal and of memory reactivation [@dimitriades2024isfs]. ...

> His other edits to this paragraph (splitting the first sentence in two, deleting "subsequently") were made against V1; V2 had already rewritten the passage — it now carries the what-is-measured distinction that

> his version does not have. Re-mapped by intent and already satisfied. Only the per-channel phrasing is a genuine preference V2 does not already express.

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§5 — NEW SECTION 2.3, and the reworked 2.4 / 2.5

2.3 Changes in sleep and EEG across aging and MCI

BEFORE: does not exist.

AFTER:

> ### 2.3 Changes in sleep and EEG across aging and MCI
>
> **[Sleep, learning and memory]** The dominant account of what sleep does for memory is systems consolidation: representations that initially depend on the hippocampus are reactivated during NREM sleep and gradually integrated into neocortical networks, where they become independent of the hippocampus and resistant to interference [@rasch2013memory]. What makes this more than a transfer of stored material is its timing. Reactivation is carried by a nested hierarchy of rhythms in which the slow oscillation groups spindles, and spindles in turn group hippocampal sharp-wave ripples, so that the information a ripple carries reaches cortex during a window in which cortical plasticity is favoured [@klinzing2019consolidation]. On this account *when* spindles occur matters as much as how many of them there are. That is the reason a rhythm which organises spindles into trains is of interest in its own right rather than as a statistical curiosity [@boutin2020spindleframework], and the expectation is borne out: how tightly spindles cluster on the infra-slow timescale predicts overnight retention [@champetier2023spindlememory].
>
> **[Sleep and aging]** Sleep changes across the lifespan in a stereotyped way. Meta-analysis of quantitative polysomnographic parameters in healthy sleepers shows total sleep time, sleep efficiency, slow-wave sleep and REM sleep all declining with age, while time in the lighter NREM stages and wake after sleep onset increase; most of the change occurs between young adulthood and middle age, after which the parameters largely plateau [@ohayon2004metaanalysis]. Clinically the picture is of sleep that is shorter, lighter and more fragmented, together with an advance of circadian phase and a blunted homeostatic response to sleep loss [@li2018normalaging]. None of this is in itself pathological, and that is exactly why it matters here: it is the baseline against which any additional effect of cognitive impairment has to be judged.
>
> **[Sleep and neurodegeneration]** The relationship between sleep and neurodegeneration runs in both directions. Disrupted sleep is a consequence of Alzheimer pathology, but it is also a contributor to it: interstitial amyloid-β rises during wakefulness and falls during sleep, and experimental sleep restriction raises it further, so that chronically poor sleep plausibly accelerates deposition — which in turn disrupts sleep [@ju2014bidirectional]. The clearance mechanism described in Section 2.1 gives that loop a physical basis [@xie2013clearance]. What makes it tractable with EEG is that the pathology tracks specific NREM features rather than sleep duration: reduced non-REM slow-wave activity is associated with tau pathology in early symptomatic Alzheimer's disease independently of how long patients sleep [@lucey2019nremtau], and the synchrony of slow waves across the scalp tracks cognitive impairment in prodromal Alzheimer's disease [@sharon2025slowwaves]. A quantitative NREM EEG measure can therefore index neurodegenerative pathology — the premise on which this thesis rests.
>
> **[Alzheimer's disease]** Alzheimer's disease is the most common cause of dementia. It is defined neuropathologically by extracellular amyloid-β plaques and intracellular neurofibrillary tangles of

hyperphosphorylated tau, accompanied by synaptic loss and neurodegeneration, and clinically by an insidious amnestic decline that later spreads to other cognitive domains [@scheltens2021alzheimer]. Its course is long, and it begins far earlier than the clinical diagnosis. Tau pathology follows an orderly anatomical progression, and in large autopsy series the earliest abnormal tau is found not in cortex but in subcortical nuclei — the locus coeruleus in particular — frequently in people in their twenties and thirties, decades before any symptom appears [@braak2011stages]. The nucleus in which that pathology appears first is also the nucleus whose activity has been linked to the infra-slow sigma rhythm in rodents, a convergence the Discussion returns to.

[Sleep in Alzheimer's disease] Sleep in Alzheimer's disease is disturbed beyond what age alone would predict. A systematic review and meta-analysis of polysomnographic studies found reduced total sleep time and sleep efficiency, more wake after sleep onset, and less slow-wave and REM sleep than in healthy older controls, with the differences consistently larger in Alzheimer's disease than in mild cognitive impairment [@zhang2022alzheimerreview]. The microstructure is affected as well, and selectively: fast spindle density is reduced over parietal cortex in both Alzheimer's disease and amnestic MCI, while frontal slow spindles are comparatively preserved [@gorgoni2016parietal].

[Mild cognitive impairment] Mild cognitive impairment (MCI) names the ground between normal aging and dementia. As originally characterised it requires a subjective cognitive complaint, objective impairment on formal testing relative to age and education, largely preserved general cognition, essentially intact activities of daily living, and the absence of dementia [@petersen1999mci]. Current criteria retain that clinical core and add an explicit second step: once the syndrome is established, the likelihood that it is *due to* Alzheimer's disease is judged separately, on aetiological rather than syndromic grounds [@albert2011mcicriteria]. The separation of the syndrome from its cause is built into the diagnosis itself. In practice identification usually begins with a brief cognitive screen, most commonly the Montreal Cognitive Assessment, which was developed specifically to detect the milder deficits that longer-established instruments miss [@nasreddine2005moca].

[Amnestic MCI — C571] MCI is subdivided by the cognitive domains affected. Amnestic MCI (aMCI), in which memory is impaired, is the subtype most closely associated with Alzheimer's disease, and that association is real: in a longitudinal clinical series, amnestic MCI progressed to probable Alzheimer's disease at 17 events per 100 person-years against 1.5 to dementia with Lewy bodies, while non-amnestic MCI showed the reverse pattern [@ferman2013nonamnestic]. It would nonetheless be wrong to treat amnestic MCI as simply an earlier phase of Alzheimer's disease. Most people diagnosed with MCI never develop dementia at all: pooling 41 robust inception cohorts, the cumulative proportion progressing to dementia was 39.2% in specialist settings and 21.9% in population samples, leading the authors to conclude that most people with MCI will not progress even after ten years of follow-up [@mitchell2009progression]. A substantial fraction moves in the opposite direction — roughly a quarter of amnestic MCI reverts to normal cognition, 14% in clinic-based samples such as the present one and 31% in community samples [@malekahmadi2016reversion]. And among those who do convert, the underlying pathology is not uniformly Alzheimer's: in an autopsy series of amnestic MCI patients who had progressed to dementia, 10 of 34 carried a non-Alzheimer primary diagnosis — hippocampal sclerosis, argyrophilic grain disease, Lewy body disease and vascular pathology — and neither demographic nor cognitive measures predicted which patients these would turn out to be [@jicha2006neuropathologic]. The position adopted here is therefore that amnestic MCI is enriched for Alzheimer's disease without being equivalent to it. A group defined by cognitive criteria is aetiologically mixed, and any disease-specific effect measured in such a group will be diluted in proportion to that mixture.

[Sleep in MCI — what is known] What is known about sleep in MCI is less settled than for Alzheimer's disease. A systematic review and meta-analysis of objectively measured sleep found that people with MCI differ from healthy older adults on polysomnographic and actigraphic measures, but with substantial

> heterogeneity between studies and effects generally smaller than those reported in dementia [@drozario2020objectivesleep]. Against that background, individual NREM features have been put forward as early markers: non-REM sleep characteristics predict subsequent cognitive impairment in aging populations [@taillard2019nonrem], and spindles, K-complexes and sleep-stage proportions have each been proposed as candidate biomarkers of amnestic MCI and Alzheimer's disease [@liu2020spindlebiomarkers]. Whether any of them separates MCI from healthy aging reliably enough to be useful remains open. That is the question this thesis puts to one particular feature of N2 sleep.

Covers his sub-topics, one paragraph each, in the order he listed them. Bracketed labels above are for your reading only and do not go into the Doc.

> **Provenance of the C571 figures.** All four are the headline numbers from the published abstracts, as recorded in `new_refs_annotated.md` §G. If you want any of them checked against the full text before it goes in, say which — the `jicha2006neuropathologic` 10-of-34 and the `ferman2013nonamnestic` event rates are the two most specific.

2.4 Changes in sleep spindles and ISFS across aging and MCI

BEFORE:

> Healthy aging reshapes NREM sleep and its spindles. Spindle density falls, slow-wave–spindle coupling loosens, and the temporal clustering of spindles breaks down, changes that themselves track cognitive decline[12–14]. In mild cognitive impairment (MCI) and Alzheimer's disease, polysomnographic studies report further sleep disruption and, in particular, reduced parietal fast-spindle density[15–16]. Because these N2 changes emerge early and carry prognostic information, sleep-EEG features have been proposed as candidate markers of incipient cognitive impairment[17–18].

AFTER — Option A (recommended, see §1.A1):

> ### 2.4 Changes in sleep spindles and ISFS across aging and MCI
>
> Healthy aging reshapes NREM sleep and its spindles. Spindle density falls, slow-wave–spindle coupling loosens, and the temporal clustering of spindles breaks down [@mander2017aging; @helfrich2018uncoupled], changes that themselves track cognitive decline [@champetier2023spindlememory]. The spindle deficits reported in amnestic MCI and Alzheimer's disease are focal rather than global: they fall on fast spindles over parietal cortex [@gorgoni2016parietal], which is the same subtype and the same region in which the ISFS is most pronounced in young adults [@lazar2019infraslow]. Because these N2 changes emerge early and carry prognostic information, sleep-EEG features have been proposed as candidate markers of incipient cognitive impairment [@taillard2019nonrem; @liu2020spindlebiomarkers].
>
> The ISFS itself, however, has been characterized almost exclusively in young adults and across development [@dimitriades2024isfs]. Only one study has examined it in a population with cognitive impairment, a concurrent report in clinically diagnosed Alzheimer's disease [@grollero2026iso], and none has asked what healthy aging on its own does to it. Whether and how the ISFS changes with healthy aging, and whether amnestic MCI adds a signal beyond that of aging, therefore remain unknown.

AFTER — Option B (minimal touch): the paragraph exactly as it stands, with `[12–14]` becoming a three-number list and `(MCI)` dropped because MCI is now defined in 2.3; then the same second paragraph as in Option A.

> **What the second paragraph is.** It is the sentence "*Yet the ISFS itself has been characterized almost exclusively in young adults; whether and how it changes with healthy aging, and whether MCI adds a signal beyond that of aging, remain unknown*", **moved here from the start of the final paragraph**, expanded by two clauses, and given the Grollero citation. It belongs under his heading — this is the "and ISFS" half of "Changes in sleep spindles **and ISFS** across aging and MCI", which the existing paragraph does not address at all. Moving it also stops 2.4 and 2.5 making the same statement twice.

2.5 The present study

BEFORE:

> Yet the ISFS itself has been characterized almost exclusively in young adults; whether and how it changes with healthy aging, and whether MCI adds a signal beyond that of aging, remain unknown. Here we quantified the ISFS during clean N2 sleep in three groups (young controls, healthy older adults, and patients with amnestic MCI), applying the analysis pipeline of Dimitriades et al. (2024) to high-density 256-channel EEG. For each participant we measured the peak frequency, bandwidth, and AUC of the fast-spindle (13–16 Hz) sigma-envelope spectrum at every channel, and compared the groups across the whole scalp, topographically, and within a pre-defined central-parietal region of interest. We asked whether these ISFS parameters and their scalp topography differ with age, and whether the ISFS of patients with amnestic MCI is comparable to, or different from, that of healthy older adults.

AFTER:

> ### 2.5 The present study
>
> Here we quantified the ISFS during N2 sleep in three groups (young controls, healthy older adults, and patients with **aMCI**), applying the analysis pipeline of Dimitriades et al. (2024) to high-density 256-channel EEG. For each participant we measured the peak frequency, bandwidth, and AUC of the fast-spindle (13–16 Hz) sigma-envelope spectrum at every **scalp** channel, and compared the groups across the whole scalp, topographically, and within a pre-defined central-parietal region of interest. We asked whether these ISFS parameters and their scalp topography differ with age, and whether the ISFS of patients with **aMCI** is comparable to, or different from, that of healthy older adults.

Opening sentence moved to 2.4. "clean" deleted, "scalp" added, "amnestic MCI" → "aMCI" — all his.

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§6 — his tracked edits, and where each one landed

6.1 Applied

#	His edit	Where
1	"occupies most of the human night" → **"occupies most of sleep in adults"**	2.2 ¶1
2	"its second stage (N2) is defined by" → **"stage 2 (N2) is characterized by the intermittent occurrence of"**	2.2 ¶1
3	"vary **systematically** in density" → "vary in density"	2.2 ¶1
4	+ **"plasticity and"** sleep-dependent memory consolidation	2.2 ¶1

#	His edit	Where
5	+ **"as well as with increased disconnection from the external sensory environment"**	2.2 ¶1
6	"In rodents" → **"In both rodents and humans"**	2.2 ¶2
7	"clocked by the locus coeruleus (LC) and the noradrenergic system" → **"driven in part by the locus coeruleus norepinephrine (LC-NE) system"**	2.2 ¶2
8	"The amplitude of **these** infra-slow noradrenergic oscillations is **in turn** coupled" → "The amplitude of infra-slow noradrenergic oscillations is coupled"	2.2 ¶2
9	"quantifying it per channel" → **"for each EEG channel"**	2.2 ¶3
10	"during **clean** N2 sleep" → "during N2 sleep"	2.5
11	"at every **scalp** channel"	2.5
12	"patients with MCI" → "patients with **aMCI**"	2.5
13	Four section headings + their sub-topic lists	the whole structure of §3 and §5

6.2 Forced by his edits or by the new sections — flagged, not silent

Change	Why
"the animal is readily aroused / the animal is relatively disconnected" → species-neutral wording	His "In both rodents and humans" makes "the animal" wrong (2.2 ¶2)
2.2 ¶1 no longer expands "non-rapid eye movement (NREM)"	Now defined in the opener and again in 2.1 ¶1; three definitions of one abbreviation
2.4 no longer expands "mild cognitive impairment (MCI)"	Now defined in 2.3
`amnestic MCI (aMCI)` defined at its first use in 2.3	Triage §8.2.4: the Introduction now needs the abbreviation, so this becomes the single body definition
Methods 3.1 must drop its definition — "patients with amnestic mild cognitive impairment (aMCI; n = 30…)" → "… impairment (n = 30…)"	Same rule — otherwise there are two body definitions. **This is a one-clause edit to already-approved Methods text.**

6.3 Superseded — V2 already does this, no action

His edit	V2 already reads
Splits ¶3's first sentence into two ("Such modulations are most prominent…")	V2 rewrote the passage and added the what-is-measured distinction his version lacks
Deletes "subsequently" from "Dimitriades et al. (2024) subsequently narrowed"	Kept — V2's sentence needs it to mark the sequence after Lázár
"whether MCI departs from healthy aging or instead mirrors it"	V2: "whether the ISFS of patients with amnestic MCI is comparable to, or different from, that of healthy older adults" — same point, and already relabelled aMCI

§7 — spotted, NOT done (scope rule: only what he asked)

1. **The opening paragraph still says the rhythm "is paced by the noradrenergic locus coeruleus".** He softened exactly this claim twice — in the Abstract ("paced by" → "associated with changes in LC-NE activity and other arousal systems and with related autonomic measures") and in 2.2 ¶2 ("clocked by" → "driven in part by"). That opening paragraph did not exist in his copy, so he never saw it. Making it consistent is a one-clause edit. **Say the word.**
2. **`galgani2023locuscoeruleus`** — LC MRI abnormality specifically in *amnestic* MCI, predicting later progression — is the strongest single citation for C571 and would fit the aMCI paragraph. `discussion_edits_before_after.md` already claims it (its ref 33). Left there to avoid using the same evidence twice.
3. **`dimitriades2024isfs` is still listed as a bioRxiv preprint** in the reference list but was published in *Sci Rep* on 18 Jun 2026 (`10.1038/s41598-026-58423-z`), and `sharon2025slowwaves` is missing its volume/issue/pages (Crossref: 21(5):e70247). Our own B1/S2 items, not his asks — but the combined numbering pass rebuilds the whole list anyway, so fixing them then costs nothing. **Your call.** *(The third of these, `andre2025remsLowing`, is moot — the Discussion pass deletes that entry.)*
4. **`bergel2026conserved`** (infra-slow sleep rhythms conserved across reptiles and mammals) would support 2.2 in one clause. Not needed for any claim he asked for, and you asked me not to overflow the citations, so it is left out.